

(Crossley §7.1) simplicial complexes

a 0-simplex is a point

a 1-simplex is a (closed, nondegenerate)
 line segment

a 2-simplex is a triangle

a 3-simplex is a tetrahedron [draw]

Df given a set S of $k + 1$ vectors
 $v_0, v_1, \dots, v_k$ in $\mathbb{R}^N$,

we say S is in general position iff $v_1 - v_0, \dots,$
 $v_k - v_0$ are linearly indep

in this case, the convex hull of S is called the
associated k -simplex $\Delta(S)$

[what is the convex hull of $v_0, v_1, \dots, v_k$?]

$$\{t_0 v_0 + t_1 v_1 + \dots + t_k v_k \\ | t_0, \dots, t_k \text{ in } [0, 1] \text{ s.t. } \sum_i t_i = 1\}$$

Df given a k -simplex $\Delta(S)$:

a subsimplex of $\Delta(S)$ is a simplex of the form

$$\Delta(S') \quad \text{for some } S' \text{ sub } S$$

if $|S'| = k - 1$, then it is called a face of $\Delta(S)$

Rem some books say “face” to mean what we
 call “subsimplex”

Df boundary of Δ = union of its faces;
interior of Δ = complement of boundary

Df a simplicial complex is a finite collection
of simplices $\{\Delta(S)\}_S$ s.t.

- 1) every point of $\bigcup_S \Delta(S)$ belongs
to the interior of $\Delta(S)$ for exactly one S
- 2) if $\Delta(S)$ is in the collection, then so is
every face of $\Delta(S)$

we say that $\bigcup_S \Delta(S)$ is the geometric realization of the simplicial complex

Ex simplicial circle [draw]
three 0-simplices
three 1-simplices
[can we use any fewer?]

Ex simplicial square [draw]
four 0-simplices
five 1-simplices
two 2-simplices

Ex simplicial annulus [draw]
six 0-simplices
twelve 1-simplices
six 2-simplices

Df a triangulation of a space X is a homeo from K to X , where K is the geometric realization of some simplicial complex

Rem an n -manifold is a second-countable, Hausdorff space X s.t. every x in X has an open nbd homeo to $\mathbb{R}^n$

(Radó '25) 2-folds have triangulations

(Moise '52) 3-folds have triangulations

(Casson '90) some 4-folds don't

(Manolescu '13) some n -folds don't,
for each $n \geq 5$

(Crossley §7.2)

Df the Euler characteristic/number of a simplicial complex is

$$e_0 - e_1 + e_2 - e_3 + \dots$$

where $e_k = \#$ of k -simplices in the collection

Ex simplicial circle: $\chi = 0$

Ex simplicial square: $\chi = 1$

Ex simplicial annulus: $\chi = 0$

Thm (Poincaré) the Euler characteristic
 only depends on the
 geometric realization K ,
 not the collection of simplices

so we sometimes denote it by $\chi(K)$

Prob show by example that by contrast,
 $e_0, e_1, \dots$ do depend on the collection

motivation: Euler's formula $V - E + F = 2$

Ex simplicial sphere: $\chi = 2$
 four 0-simplices
 six 1-simplices
 four 2-simplices

Ex [Example 7.8 in Crossley]
 simplicial torus: $\chi = 0$
 nine 0-simplices
 twenty-seven 1-simplices
 eighteen 2-simplices